

Cryptography Mechanisms

For Access Control In Wireless Sensor Networks

For secure communication among sensors, secret keys must be established between them. Many key establishment techniques have been designed to address the trade-off between limited memory and security, but which scheme is the most effective is still debatable

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Security allows wireless sensor networks (WSNs) to be used with confidence. Without security, the use of WSN in any application domain would result in undesirable consequences.

The basic idea of a WSN is to disperse tiny sensing devices that are capable of sensing some changes of incidents/parameters and communicating with other devices, over a specific geographic area for some specific purposes like target tracking, surveillance and environmental monitoring.

Basically, the major challenge for employing any efficient security scheme in wireless sensor networks is created by the size of sensors, followed by the processing power, memory and type of tasks expected from the sensors.

Challenges in providing security to a WSN

First, there is a conflicting interest between minimisation of resource consumption and maximisation of security level. During the design of any security solution, one needs to take care of limited energy, limited memory, limited computing power, limited communication bandwidth and limited communication range.

Second, the type of security mechanism that can be hosted on a sensor node platform is dependent on the capabilities and constraints of sensor node hardware.

Third, the ad-hoc networking topology of WSN facilitates attackers for different types of link attacks ranging

from passive eavesdropping to active interference. Attacks on a WSN can come from all directions and target any node leading to leakage of secret information, interfering message, impersonating nodes, etc.

Fourth, the communication in WSN is through wireless media, mainly radio. This characteristic of WSN makes wire-based security scheme impractical for a WSN.

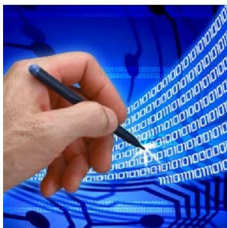
Fifth, the topology of WSN is always dynamic. The sensor nodes can come and go in an arbitrary fashion. Node failures may be permanent or intermittent, and this gives a higher level of system dynamics. Besides, very often large numbers of nodes are expected in sensor network deployments and the nature of this deployment is unpredictable.

Finally, the overall cost of the WSN should be as low as possible.

Basic security schemes in WSNs

Security is a broadly used term encompassing the characteristics of authentication, integrity, privacy, non-repudiation and anti-playback. Every eligible receiver should receive all messages intended for it and be able to verify the integrity of every message as well as the identity of the sender. Adversaries should not be able to infer the contents of any message.

In conventional computer networks, the primary security goal is reliable delivery of messages (protection against denial-of-service (DoS) attack). Message authenticity, integrity and confidentiality are usually achieved



by an end-to-end security mechanism such as secure socket layer. This is because the dominating traffic pattern is end-to-end communication, where it is neither necessary nor desirable for the contents of the message (beyond the necessary headers) to be available to the intermediate routers. The more the dependency on the information provided by the networks, the higher the risk to secure transmission of information over the networks. For secure transmission of various types of information over networks, several cryptographic, steganographic and other techniques are needed.

Since sensor nodes use wireless communications, eavesdropping, injection, replay and other attacks can be placed on the network. The adversary is able to deploy malicious nodes in the network or compromise some legitimate nodes.

Cryptography. The encryption-decryption techniques devised for the traditional wired networks are not feasible to be applied directly to wireless